

Web Programming Laboratory

CSE 3100

PROJECT REPORT

Title: Vehicle Rental Management System
(DriveEase)

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1. Project Overview

1.1 Purpose

The Vehicle Rental Management System (DriveEase) is a comprehensive web based platform designed to streamline and automate the operations of a vehicle rental business. The system enables administrators to manage their fleet of vehicles, process bookings, track payments and generate reports, while providing customers with an intuitive interface to browse, compare and rent vehicles.

1.2 Objectives

- Develop a centralized platform for managing vehicle rental operations.
- Provide an intuitive customer facing interface for browsing and booking vehicles.
- Implement role based access control for administrators and customers.
- Enable real time vehicle availability tracking and booking conflict detection.
- Generate comprehensive business reports for revenue, fleet performance and customer analytics.
- Support multiple payment methods and automated payment tracking.
- Maintain a complete audit trail through notifications and booking history.

1.3 Problem Statement

Traditional vehicle rental businesses often rely on manual processes such as paper based booking records, spreadsheets for fleet management and phone based communication. These methods are error prone, time consuming, and do not scale well. Customers face difficulties in comparing vehicles, checking availability and making informed decisions. There is a need for a digital solution that automates the entire rental workflow, from vehicle browsing to booking confirmation, payment processing and report generation.

1.4 Scope

- Public vehicle browsing with search, filter, and comparison capabilities.
- User registration and authentication with role based access.
- Customer dashboard with booking management, wishlist and profile.
- Admin dashboard with fleet management, booking approval and analytics.
- Vehicle CRUD operations with multi image support.
- Booking system with date overlap conflict detection.
- Payment tracking with multiple payment method support
- Maintenance scheduling and cost tracking.
- Review and rating system with admin moderation.
- PDF report generation for revenue, vehicles, and customer analytics.
- Notification system for real time updates.

1.5 Target Users

User Role	Description	Key Capabilities
Administrator	System administrator managing the rental business	Fleet management, booking approval, payment processing, report generation, customer management
Customer	End-user renting vehicles	Browse vehicles, create bookings, manage wishlist, view invoices, submit reviews, manage profile
Guest	Unauthenticated visitor	Browse vehicles, view details, compare vehicles, read FAQ/About pages

2. Project Description

DriveEase is a full stack web application built on the Laravel 12 framework that provides a complete solution for vehicle rental management. The system follows the Model-View-Controller (MVC) architecture pattern and uses SQL queries for database operations, providing fine-grained control over data retrieval and manipulation.

2.1 What the System Does

The system serves as a complete rental management platform with three distinct interfaces: a public-facing website for vehicle browsing, a customer portal for booking management and an administrative panel for business operations. It handles the entire lifecycle of a vehicle rental from initial browsing through booking, payment and post rental review.

2.2 Why It Was Developed

The system was developed to address the challenges faced by vehicle rental businesses in managing their operations efficiently. It eliminates manual processes, reduces errors, and provides a modern, accessible platform for both business owners and customers.

2.3 Real-World Use Cases

- A customer needs to rent a car for a weekend trip and uses the platform to compare available vehicles by price, features and reviews.
- An administrator adds new vehicles to the fleet, sets pricing, and manages maintenance schedules.
- A customer submits a booking request and receives real-time notifications about booking status changes.
- The business owner generates monthly revenue reports to analyze business performance.
- An administrator reviews and moderates customer reviews to maintain quality standards.

2.4 Expected Benefits

- Reduced manual effort in managing bookings and fleet operations.
- Improved customer experience through self-service vehicle browsing and booking.
- Better business insights through comprehensive reporting and analytics.
- Enhanced accountability through complete audit trails and notification system.
- Scalable architecture that can grow with the business.

3. Technology Used

3.1 Backend Technologies

Technology	Version	Purpose	Role in Project
PHP	^8.2	Server-side scripting language	Core runtime for the Laravel framework, handles all server-side logic
Laravel	12.x	PHP web application framework	Provides MVC architecture, routing, authentication, database abstraction, and templating
MySQL	8.x	Relational database management system	Stores all application data including users, vehicles, bookings, and payments
Composer	^2.x	PHP dependency manager	Manages backend packages and autoloading configuration

3.2 Frontend Technologies

Technology	Version	Purpose	Role in Project
HTML5	-	Markup language	Structures all web pages and forms
CSS3	-	Styling language	Provides visual styling, animations, and responsive design
Bootstrap	5.3.2	CSS framework	Provides responsive grid system, components, and utilities (loaded via CDN)
JavaScript	ES6+	Programming language	Handles client-side interactivity, AJAX requests, and dynamic UI updates
jQuery	3.7.1	JavaScript library	Simplifies DOM manipulation, event handling, and AJAX calls (CDN)
Vite	7.x	Frontend build tool	Bundles and optimizes frontend assets (CSS, JS)
Tailwind CSS	4.x	Utility-first CSS	Available via Vite plugin

		framework	for utility classes
Axios	1.11.0	HTTP client	Handles AJAX requests from the frontend

3.3 Supporting Technologies

Technology	Version	Purpose	Role in Project
Font Awesome	6.5.0	Icon library	Provides scalable vector icons throughout the UI (CDN)
SweetAlert2	-	JavaScript alert library	Provides beautiful, customizable alert dialogs and toast notifications
Chart.js	4.4.1	JavaScript charting library	Renders interactive charts for admin dashboard analytics (CDN)
DataTables	1.10.21	jQuery table plugin	Enhances HTML tables with search, sort, and pagination (CDN)
Animate.css	4.1.1	CSS animation library	Provides smooth CSS animations for UI elements (CDN)
DomPDF	-	PDF generation library	Generates PDF reports for revenue and vehicle analytics
Nominatim API	-	Geolocation API	OpenStreetMap reverse geocoding for pickup location detection

4. System Architecture

4.1 MVC Architecture

The application follows the Model-View-Controller (MVC) architectural pattern provided by the Laravel framework. This pattern separates the application into three interconnected components:

Models: The application uses a single Eloquent model (User) and raw SQL queries via the DB facade for all other database operations. This provides direct control over SQL execution and query optimization.

Views: Blade templating engine renders all UI components with 48 Blade templates organized across three layout systems (public, admin, customer).

Controllers: 18 functional controllers handle HTTP requests, execute business logic using SQL queries, and return appropriate views or responses.

4.2 Routing System

All routes are defined in routes/web.php. The application uses 70 routes organized into three groups: public routes (17), customer routes (13), and admin routes (37). Route groups use middleware for authentication and authorization, with URL prefixes and name prefixes for organization.

4.3 Middleware

The application uses two custom middleware classes in addition to Laravel's built-in auth middleware:

Middleware	File	Purpose
auth	Built-in	Ensures user is authenticated before accessing protected routes
AdminMiddleware	app/Http/Middleware/AdminMiddleware.php	Restricts access to admin users only by checking role === "admin"
CustomerMiddleware	app/Http/Middleware/CustomerMiddleware.php	Restricts access to customer users only by checking role === "customer"

5. Project Folder Structure

app/ Contains the core application code including Controllers, Models, Middleware, Providers, and Support files.

app/Http/Controllers/ All 18 controllers organized by namespace: Root (HomeController, VehicleController), Auth (LoginController, RegisterController), Admin (7 controllers), Customer (5 controllers).

app/Models/ Contains the single Eloquent model: User.php.

app/Http/Middleware/ Custom middleware: AdminMiddleware and CustomerMiddleware.

app/Providers/ Service providers including AppServiceProvider (registers NotificationComposer).

app/Support/ Helper functions (helpers.php) for status badges, image URLs, and star ratings.

bootstrap/ Application bootstrapping and cache configuration.

config/ Configuration files for database, session, cache, mail, filesystems, and other services.

database/ Contains migrations (6 files creating 16 tables), seeders (7 files), and factories.

database/migrations/ Database schema definitions for all tables.

database/seeders/ Data seeders: UserSeeder, VehicleCategorySeeder, VehicleSeeder, BookingSeeder, MaintenanceSeeder, ReviewSeeder.

public/ Web-accessible root directory with index.php, CSS, JavaScript, and uploaded files (vehicles, categories, avatars).

resources/ Views, CSS, JavaScript source files, and language files.

resources/views/ 48 Blade templates organized: layouts/ (3), partials/ (2), auth/ (2), home/ (9), admin/ (21), customer/ (7).

routes/ Route definitions: web.php (70 routes), console.php.

storage/ Application storage: logs, compiled views, cache, and user-uploaded files.

tests/ PHPUnit test files (not implemented in current version).

vendor/ Composer-managed third-party packages and autoloader.

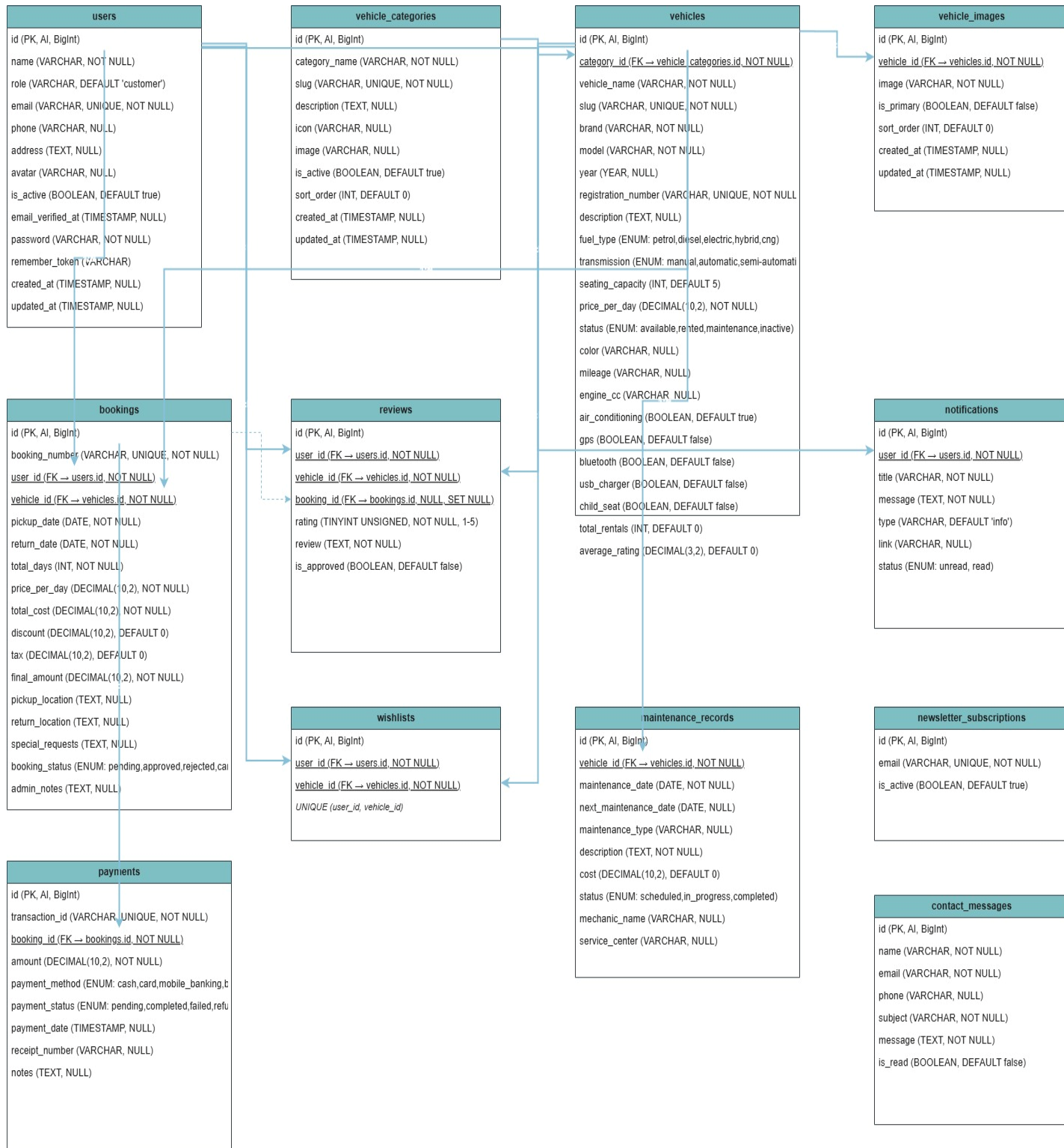
6. Database Design

6.1 Entity Relationships

The database consists of 16 tables with 13 foreign key constraints. The core entities are Users, Vehicles, Bookings, and Payments, with supporting tables for categories, images, reviews, wishlists, maintenance records, notifications, and communication.

Relationship	Type	Description
users → bookings	1:N	A user can have multiple bookings
users → reviews	1:N	A user can write multiple reviews
users → wishlists	1:N	A user can wishlist multiple vehicles
users → notifications	1:N	A user can receive multiple notifications
vehicle_categories → vehicles	1:N	A category contains multiple vehicles
vehicles → vehicle_images	1:N	A vehicle can have multiple images
vehicles → bookings	1:N	A vehicle can be booked multiple times
vehicles → reviews	1:N	A vehicle can receive multiple reviews
vehicles → maintenance_records	1:N	A vehicle can have multiple maintenance records
bookings → payments	1:1	A booking has one associated payment
bookings → reviews	1:N	A booking can have a review (nullable FK)

6.2 Schema Diagram



7. Authentication & Authorization

7.1 Authentication

The application implements session-based authentication using Laravel's built-in Auth system. Users authenticate via email and password, with passwords hashed using bcrypt.

Feature	Implementation
Login	POST /login → LoginController::login() validates credentials, checks is_active, calls Auth::login()
Registration	POST /register → RegisterController::register() creates user with "customer" role and hashed password
Logout	POST /logout → LoginController::logout() calls Auth::logout(), invalidates session, regenerates CSRF token
Remember Me	Optional "remember me" checkbox on login form sets remember token cookie
Session Storage	Database-backed sessions (SESSION_DRIVER=database) for persistence across server restarts
CSRF Protection	Laravel's built-in CSRF token verification on all POST/PUT/DELETE forms

7.2 Authorization

Authorization is implemented through a role-based access control system using two custom middleware classes.

Role	Middleware	Access Level
admin	AdminMiddleware	Full access to admin panel, fleet management, booking approval, payment processing, reports, and customer management
customer	CustomerMiddleware	Access to customer dashboard, booking creation/management, wishlist, profile, and reviews
guest	None (public)	Access to public pages: home, vehicle listings, vehicle details, about, contact, FAQ, login,

		registration
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7.3 Role Checks

The User model implements two role-check methods used by middleware and controllers:

```
public function isAdmin(): bool { return $this->role === "admin"; }
public function isCustomer(): bool { return $this->role === "customer"; }
```

8. Project Features

Feature	Description	Controller/Method	Database Tables
Public Vehicle Browsing	Browse 45 vehicles across 8 categories with search, filter (fuel type, transmission, price range), sort, and pagination (12/page).	VehicleController	vehicles, vehicle_images, vehicle_categories
Vehicle Detail Page	View complete vehicle information including images, specifications, accessories, reviews, and related vehicles.	VehicleController::show	vehicles, vehicle_images, reviews
Vehicle Comparison	Compare up to 4 vehicles side-by-side on specifications, pricing, and features.	VehicleController::compare	vehicles, vehicle_images
Live Search	AJAX-powered search with instant results dropdown in the navigation bar.	VehicleController::search	vehicles
User Registration	Customer registration with name, email, phone, and password confirmation.	RegisterController	users
User Login/Logout	Session-based authentication with role-based redirection.	LoginController	users, sessions
Customer Dashboard	Overview of bookings, notifications, and quick actions.	Customer\DashboardController	bookings, notifications
Booking Creation	Date-overlap detection, cost calculation, and multi-step booking process.	Customer\BookingController	bookings, payments, notifications
Booking Management	View, cancel, and track booking status with real-	Customer\BookingController	bookings, notifications

	time notifications.		
Invoice Generation	PDF-ready invoice view with booking and payment details.	Customer\BookingController::invoice	bookings, payments
Wishlist	Add/remove vehicles to personal wishlist with AJAX toggle.	Customer\WishlistController	wishlists
Profile Management	Update personal information, avatar, and password.	Customer\ProfileController	users
Review System	Submit reviews (1-5 stars) for completed bookings, pending admin approval.	Customer\ReviewController	reviews
Admin Dashboard	10 key metrics, revenue chart, booking distribution, top vehicles, recent activity.	Admin\DashboardController	multiple tables
Vehicle CRUD	Full create/read/update/delete with multi-image upload and slug generation.	Admin\VehicleController	vehicles, vehicle_images
Category CRUD	Manage 8 vehicle categories with icons and images.	Admin\CategoryController	vehicle_categories
Booking Approval	Approve/reject/complete bookings with transaction-wrapped status updates.	Admin\BookingController	bookings, payments, vehicles
Payment Management	Track and update payment statuses with audit trail.	Admin\PaymentController	payments
Customer Management	View customer list, details, toggle active status.	Admin\CustomerController	users, bookings
Maintenance Tracking	Schedule and track vehicle maintenance with cost tracking.	Admin\MaintenanceController	maintenance_records, vehicles
Review Moderation	Approve/reject/delete customer reviews with rating recalculation.	Admin\ReviewController	reviews, vehicles
Reports & PDF Export	Revenue, vehicle performance, customer, and maintenance reports with PDF export.	Admin\ReportController	multiple tables
Geolocation API	Auto-detect pickup location using browser geolocation and	JavaScript (client-side)	-

	Nominatim reverse geocoding.		
Newsletter Subscription	Email newsletter subscription via homepage form.	HomeController::newsletter	newsletter_subscriptions
Contact Form	Contact form submission with admin notification.	HomeController::contactSubmit	contact_messages
Notifications	Real-time notification system with bell icon and badge counts.	NotificationComposer	notifications
Responsive Design	Mobile-friendly layouts using Bootstrap 5.3.2 grid system.	All Blade views	-

9. Route Analysis

9.1 Public Routes

Method	URI	Controller	Middleware	Description
GET	/	HomeController@index	None	Home page
GET	/about	HomeController@about	None	About page
GET	/contact	HomeController@contact	None	Contact form
POST	/contact	HomeController@contactSubmit	None	Submit contact form
GET	/faq	HomeController@faq	None	FAQ page
GET	/privacy	HomeController@privacy	None	Privacy policy
GET	/terms	HomeController@terms	None	Terms of service
POST	/newsletter	HomeController@newsletter	None	Subscribe to newsletter
GET	/vehicles	VehicleController@index	None	Vehicle listing
GET	/vehicles/compare	VehicleController@compare	None	Compare vehicles
GET	/vehicles/search	VehicleController@search	None	AJAX search
GET	/vehicles/{slug}	VehicleController@show	None	Vehicle detail
GET	/login	LoginController@showForm	None	Login form
POST	/login	LoginController@login	None	Process login
POST	/logout	LoginController@logout	None	Logout
GET	/register	RegisterController@showForm	None	Registration form

POST	/register	RegisterController@register	None	Process registration
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9.2 Customer Routes

Method	URI	Controller	Middlewar e	Descriptio n
GET	/customer/dashboard	Customer\DashboardController@index	auth, customer	Customer dashboard
GET	/customer/bookings	Customer\BookingController@index	auth, customer	My bookings list
GET	/customer/bookings/create/{slug}	Customer\BookingController@create	auth, customer	Create booking form
POST	/customer/bookings	Customer\BookingController@store	auth, customer	Submit booking
GET	/customer/bookings/{id}	Customer\BookingController@show	auth, customer	Booking detail
POST	/customer/bookings/{id}/cancel	Customer\BookingController@cancel	auth, customer	Cancel booking
GET	/customer/bookings/{id}/invoice	Customer\BookingController@invoice	auth, customer	View invoice
GET	/customer/wishlist	Customer\WishlistController@index	auth, customer	My wishlist
POST	/customer/wishlist/toggle	Customer\WishlistController@toggle	auth, customer	Toggle wishlist
GET	/customer/profile	Customer\ProfileController@edit	auth, customer	Edit profile
POST/PUT	/customer/profile	Customer\ProfileController@update	auth, customer	Update profile
POST	/customer/password	Customer\ProfileController@changePassword	auth, customer	Change password
POST	/customer/reviews	Customer\ReviewController@store	auth, customer	Submit review

9.3 Admin Routes

Method	URI	Controller	Middlewar e	Description
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GET	/admin/dashboard	Admin\DashboardController@index	auth, admin	Admin dashboard
GET	/admin/vehicles	Admin\VehicleController@index	auth, admin	Vehicle list
GET	/admin/vehicles/create	Admin\VehicleController@create	auth, admin	Create vehicle form
POST	/admin/vehicles	Admin\VehicleController@store	auth, admin	Store vehicle
GET	/admin/vehicles/{id}	Admin\VehicleController@show	auth, admin	Vehicle detail
GET	/admin/vehicles/{id}/edit	Admin\VehicleController@edit	auth, admin	Edit vehicle form
PUT	/admin/vehicles/{id}	Admin\VehicleController@update	auth, admin	Update vehicle
DELETE	/admin/vehicles/{id}	Admin\VehicleController@destroy	auth, admin	Delete vehicle
GET	/admin/categories	Admin\CategoryController@index	auth, admin	Category list
GET	/admin/categories/create	Admin\CategoryController@create	auth, admin	Create category form
POST	/admin/categories	Admin\CategoryController@store	auth, admin	Store category
GET	/admin/categories/{id}/edit	Admin\CategoryController@edit	auth, admin	Edit category form
PUT	/admin/categories/{id}	Admin\CategoryController@update	auth, admin	Update category
DELETE	/admin/categories/{id}	Admin\CategoryController@destroy	auth, admin	Delete category
GET	/admin/bookings	Admin\BookingController@index	auth, admin	Booking list
GET	/admin/bookings/{id}	Admin\BookingController@show	auth, admin	Booking detail
POST	/admin/bookings/{id}/approve	Admin\BookingController@approve	auth, admin	Approve booking
POST	/admin/bookings/{id}/reject	Admin\BookingController@reject	auth, admin	Reject booking
POST	/admin/bookings/{id}/complete	Admin\BookingController@complete	auth, admin	Complete booking
GET	/admin/payments	Admin\PaymentController@index	auth,	Payment list

			admin	
GET	/admin/payments/{id}	Admin\PaymentController@show	auth, admin	Payment detail
POST	/admin/payments/{id}/status	Admin\PaymentController@updateStatus	auth, admin	Update payment status
GET	/admin/customers	Admin\CustomerController@index	auth, admin	Customer list
GET	/admin/customers/{id}	Admin\CustomerController@show	auth, admin	Customer detail
POST	/admin/customers/{id}/toggle	Admin\CustomerController@toggleStatus	auth, admin	Toggle customer status
GET	/admin/maintenance	Admin\MaintenanceController@index	auth, admin	Maintenance list
GET	/admin/maintenance/create	Admin\MaintenanceController@create	auth, admin	Create maintenance form
POST	/admin/maintenance	Admin\MaintenanceController@store	auth, admin	Store maintenance record
GET	/admin/maintenance/{id}/edit	Admin\MaintenanceController@edit	auth, admin	Edit maintenance form
PUT	/admin/maintenance/{id}	Admin\MaintenanceController@update	auth, admin	Update maintenance record
DELETE	/admin/maintenance/{id}	Admin\MaintenanceController@destroy	auth, admin	Delete maintenance record
GET	/admin/reviews	Admin\ReviewController@index	auth, admin	Review list
POST	/admin/reviews/{id}/approve	Admin\ReviewController@approve	auth, admin	Approve review
POST	/admin/reviews/{id}/reject	Admin\ReviewController@reject	auth, admin	Reject review
DELETE	/admin/reviews/{id}	Admin\ReviewController@destroy	auth, admin	Delete review
GET	/admin/reports	Admin\ReportController@index	auth, admin	Reports landing
GET	/admin/reports/revenue	Admin\ReportController@revenue	auth, admin	Revenue report

GET	/admin/reports/vehicles	Admin\ReportController@vehicles	auth, admin	Vehicle report
GET	/admin/reports/customers	Admin\ReportController@customers	auth, admin	Customer report
GET	/admin/reports/maintenance	Admin\ReportController@maintenance	auth, admin	Maintenanc e report

10.Middleware

Middleware	Type	Purpose	Logic
auth	Built-in	Authentication guard	Checks if user is logged in via session. Redirects unauthenticated users to login page.
AdminMiddleware	Custom	Admin authorization	Verifies auth()->check() and auth()->user()->isAdmin(). Aborts with 403 if not admin.
CustomerMiddleware	Custom	Customer authorization	Verifies auth()->check() and auth()->user()->isCustomer(). Aborts with 403 if not customer.
VerifyCsrfToken	Built-in	CSRF protection	Verifies CSRF token on POST/PUT/DELETE requests. Excepts specific URIs if configured.
EncryptCookies	Built-in	Cookie encryption	Encrypts all cookies for security.
TrimStrings	Built-in	Input sanitization	Trims whitespace from all request input.
ConvertEmptyStringsToNull	Built-in	Input sanitization	Converts empty string inputs to null for database storage.

11. Security Features

11.1 CSRF Protection

All forms include Laravel's CSRF token (@csrf directive). The VerifyCsrfToken middleware validates tokens on all state-changing requests (POST, PUT, DELETE). This prevents Cross-Site Request Forgery attacks.

11.2 SQL Injection Prevention

All database queries use parameterized statements with bound parameters via DB::select(), DB::insert(), DB::update(). User input is never directly concatenated into SQL strings.

11.3 Password Hashing

Passwords are hashed using bcrypt (via Hash::make()) before storage. The User model casts the password field to "hashed" for automatic hashing. Login verification uses Hash::check().

11.4 XSS Prevention

Blade templating engine automatically escapes HTML output using htmlspecialchars(). User-supplied content is rendered safely in views.

11.5 Authentication

Session-based authentication with database-backed sessions. CSRF protection on all forms. "Remember me" functionality with encrypted cookies.

11.6 Authorization

Role-based access control using custom middleware (AdminMiddleware, CustomerMiddleware). Route groups enforce role requirements. Controller methods verify ownership for customer-specific operations.

11.7 Input Validation

Server-side validation on all form submissions using Laravel's Validator. Rules include required fields, type checking, length constraints, and database uniqueness checks.

11.8 Session Security

Database-backed sessions (SESSION_DRIVER=database) for persistence. Session invalidation on logout. CSRF token regeneration on login.

12. Version Control:

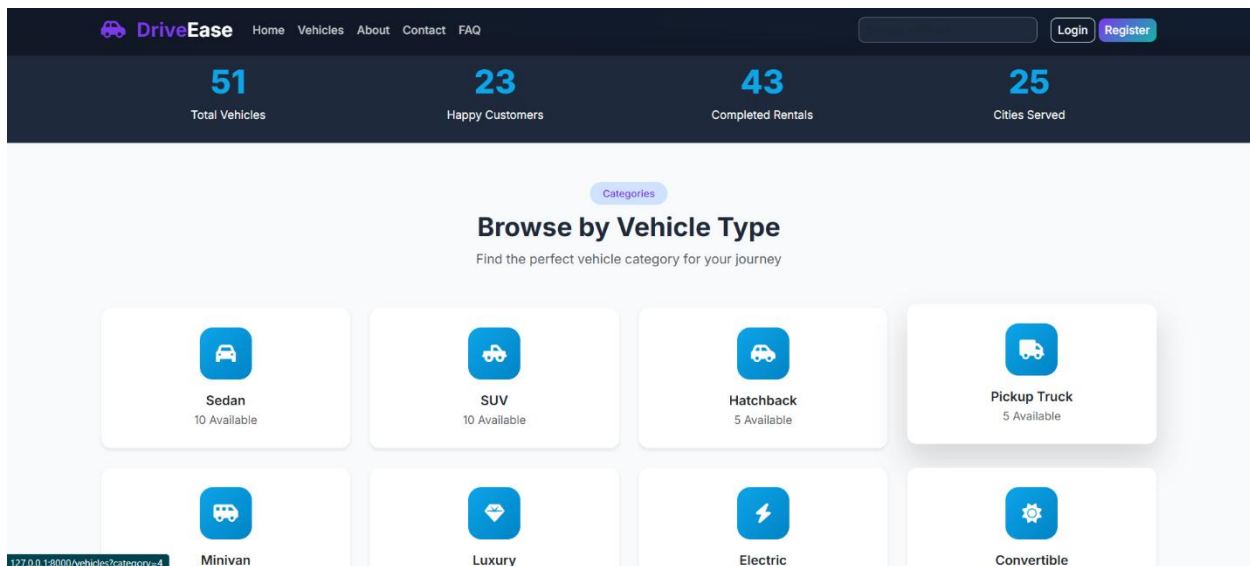
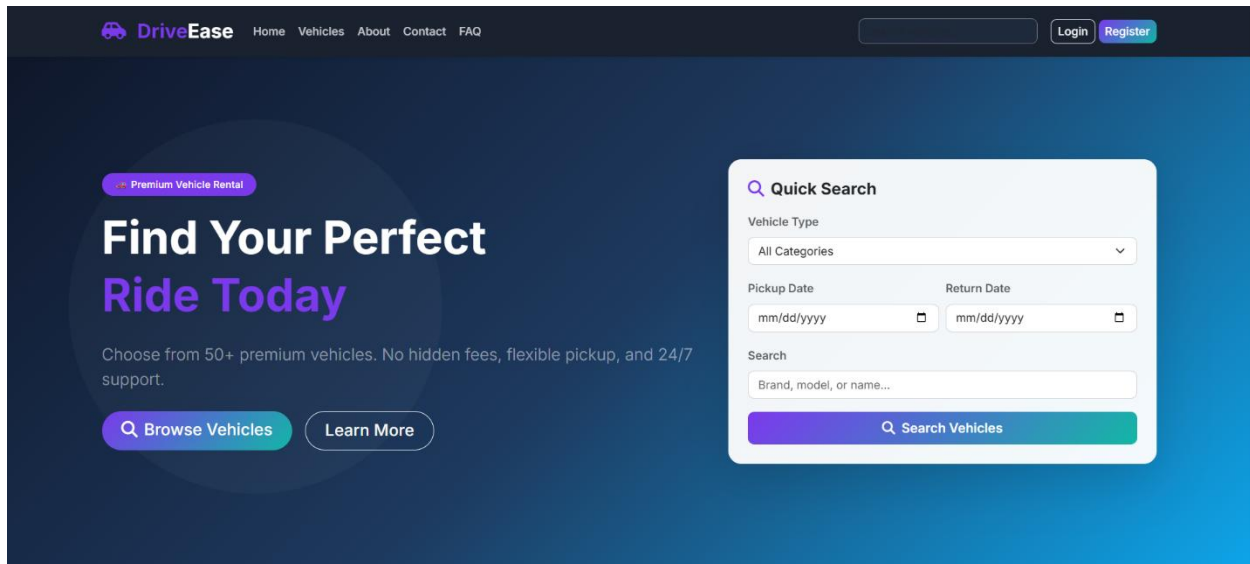
Commits on Jul 7, 2026	Add view composer and public CSS assets shahriarhossainprottoy committed 2 weeks ago	51f85eb		
Commits on Jul 5, 2026	Add database seeders with sample data shahriarhossainprottoy committed 2 weeks ago	89a12d6		
Commits on Jul 4, 2026	Create customer dashboard and booking system shahriarhossainprottoy committed 2 weeks ago	7e31dd5		
Commits on Jul 2, 2026	Implement admin maintenance, reviews and reports shahriarhossainprottoy committed 3 weeks ago	8fb75e8		
Commits on Jul 1, 2026	Add admin booking, customer and payment modules shahriarhossainprottoy committed 3 weeks ago	c70ee05		
Commits on Jun 29, 2026	Implement admin dashboard and vehicle management shahriarhossainprottoy committed 3 weeks ago	5885bec		

Commits on Jul 13, 2026	Update vehicle image mappings to use model-specific photos shahriarhossainprottoy committed 5 days ago	bf59fbe		
Commits on Jul 14, 2026	Add User model role methods and contact form error display shahriarhossainprottoy committed last week	5168137		
Commits on Jul 12, 2026	Fix layout visibility, footer and font styling shahriarhossainprottoy committed last week	988c9e1		
Commits on Jul 11, 2026	Update seeders with model-specific vehicle images shahriarhossainprottoy committed last week	6a540c7		
Commits on Jul 10, 2026	Add route definitions, middleware and helpers shahriarhossainprottoy committed last week	f3c6a9d		
Commits on Jul 8, 2026	Fix database schema and migration issues shahriarhossainprottoy committed 2 weeks ago	5dd15a9		

Commits on Jul 19, 2026	Support PUT method for profile update route and clean up comments shahriarhossainprottoy committed 14 hours ago	eae784		
	Add car favicon and emoji to browser tab titles shahriarhossainprottoy committed 18 hours ago	7850df8		
Commits on Jul 18, 2026	Add geolocation API for pickup location auto-detection shahriarhossainprottoy committed 2 days ago	ed90876		
Commits on Jul 17, 2026	Update currency and styling across customer and home views shahriarhossainprottoy committed 3 days ago	c1a53a3		
Commits on Jul 16, 2026	Update currency from dollar to Taka across admin views shahriarhossainprottoy committed 4 days ago	081cc08		
Commits on Jul 15, 2026	Update vehicle image mappings to use model-specific photos shahriarhossainprottoy committed 5 days ago	bf59fbe		

13.Screenshots:

Landing Page:



Vehicle Page:

 DriveEase

[Home](#) [Vehicles](#) [About](#) [Contact](#) [FAQ](#)

[Login](#) [Register](#)

Most Popular Vehicles

[View All](#)

Available



Mazda 6 Sedan
5 Petrol Automatic
★★★★★ (0)
Tk 7,200 /day [Book Now](#)

Available



Ford F-150 Pickup Truck
5 Petrol Automatic
★★★★★ (1)
Tk 15,000 /day [Book Now](#)

Available



Kia Sportage SUV
5 Petrol Automatic
★★★★★ (0)
Tk 9,500 /day [Book Now](#)

Available



Honda Odyssey Minivan
8 Petrol Automatic
★★★★★ (0)
Tk 13,500 /day [Book Now](#)

Available



Chevrolet Equinox SUV

Available



Toyota Yaris Hatchback

Available



Mazda CX-5 SUV

Available



Honda Civic Sedan

 DriveEase

[Home](#) [Vehicles](#) [About](#) [Contact](#) [FAQ](#)

   Shahriar Hossain

Filters [Clear all](#)

SEARCH

CATEGORY

- ☐ Sedan (10)
- ☐ SUV (10)
- ☐ Hatchback (5)
- ☐ Pickup Truck (5)
- ☐ Minivan (3)
- ☒ Luxury (7)
- ☐ Electric (6)
- ☐ Convertible (5)

FUEL TYPE

- ☐ Petrol
- ☐ Diesel
- ☐ Electric
- ☐ Hybrid
- ☐ Cng

TRANSMISSION

Available



Mercedes C-Class Luxury
Mercedes C-Class 2023
5 Petrol Automatic
★★★★★ (2)
Tk 22,000 /day [View Details](#)

Available



BMW 3 Series Luxury
BMW 3 Series 2021
5 Petrol Automatic
★★★★★ (2)
Tk 18,000 /day [View Details](#)

Available



Audi A6 Luxury
Audi A6 2022
5 Petrol Automatic
★★★★★ (0)
Tk 21,000 /day [View Details](#)

Available



Mercedes C-Class Luxury

Available



Nissan Leaf Electric

Available



Audi A6 Luxury

 DriveEase

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FUEL TYPE

- ☐ Petrol
- ☐ Diesel
- ☐ Electric
- ☐ Hybrid
- ☐ Cng

TRANSMISSION

- ☐ Manual
- ☐ Automatic
- ☐ Semi-automatic

PRICE PER DAY
Tk 0 **Tk 25000**

[Apply Filters](#)

Available



Mercedes C-Class Luxury
Mercedes C-Class 2023
5 Petrol Automatic
★★★★★ (2)
Tk 22,000 /day [View Details](#)

Available



Nissan Leaf Electric
Nissan Leaf 2022
5 Electric Automatic
★★★★★ (0)
Tk 9,000 /day [View Details](#)

Available



Toyota Land Cruiser SUV
Toyota Land Cruiser 2022
8 Diesel Automatic
★★★★★ (1)
Tk 20,000 /day [View Details](#)

1

2

3

4

5

User Dashboard

 DriveEase

[Home](#)

[Vehicles](#)

SH

Shahriar Hossain

prottoy@gmail.com

Dashboard

My Bookings

Wishlist

Profile

Browse Vehicles

Logout

Welcome back, Shahriar Hossain! 🍌

Here's a summary of your rental activity.

2

Total Bookings

2

Active Rentals

0

Completed

0

Pending

0

Wishlist

4

Notifications

Recent Bookings

View All



Tesla Model 3

Jul 28 → Jul 29, 2026 · 1 days

Tk 16,800

Approved



Chrysler Pacifica

Jul 20 → Jul 21, 2026 · 1 days

Tk 15,225

Approved

 DriveEase

[Home](#)

[Vehicles](#)

SH

Shahriar Hossain

prottoy@gmail.com

Dashboard

My Bookings

Wishlist

Profile

Browse Vehicles

Logout

My Bookings (2)

+ New Booking



Tesla Model 3

BK-RGHQSIU1

Jul 28, 2026 → Jul 29, 2026 · 1 days · Tk 16,800.00 · Cash

Approved

Details

Invoice

Cancel



Chrysler Pacifica

BK-F3E3A0BL

Jul 20, 2026 → Jul 21, 2026 · 1 days · Tk 15,225.00 · Cash

Approved

Details

Invoice

Cancel

Admin Dashboard:

DriveEase

Admin

MAIN

Dashboard

FLEET

Vehicles

Categories

Maintenance

RENTALS

Bookings

Payments

PEOPLE

Customers

Reviews

ANALYTICS

Reports

ACCOUNT

Admin / Dashboard

51

Total Vehicles

49

Available

23

Total Customers

103

Total Bookings

13

Pending Bookings

29

Active Rentals

43

Completed

Tk 1,834,980

Monthly Revenue

Monthly Revenue (Last 12 Months)

Total: Tk 6,855,240

Booking Status

DriveEase

Admin

MAIN

Dashboard

FLEET

Vehicles

Categories

Maintenance

RENTALS

Bookings

Payments

PEOPLE

Customers

Reviews

ANALYTICS

Reports

ACCOUNT

Admin / Vehicles

Vehicle Management

+ Add Vehicle

Search name, brand, reg...

All Categories

All Status

Filter

Clear

#	Vehicle	Category	Reg. No.	Fuel / Trans.	Price/Day	Status	Rentals	Actions
51	Toyota Land Cruiser Toyota Land Cruiser 2022	SUV	DHA-1051	Diesel / Automatic	Tk 20,000.00	Available	61	
50	Mitsubishi Outlander Mitsubishi Outlander 2022	SUV	DHA-1050	Petrol / Automatic	Tk 11,500.00	Available	29	
49	Kia Cerato Kia Cerato 2021	Sedan	DHA-1049	Petrol / Automatic	Tk 6,000.00	Available	69	
47	Toyota Corolla Toyota Corolla 2021	Sedan	DHA-1047	Petrol / Automatic	Tk 6,200.00	Available	25	
48	Honda Civic Honda Civic 2022	Sedan	DHA-1048	Petrol / Automatic	Tk 6,500.00	Available	70	
46	Porsche Boxster Porsche Boxster 2022	Convertible	DHA-1046	Petrol / Automatic	Tk 25,000.00	Available	32	

DriveEase

Admin

MAIN

Dashboard

FLEET

Vehicles

Categories

Maintenance

RENTALS

Bookings

Payments

PEOPLE

Customers

Reviews

ANALYTICS

Reports

ACCOUNT

Admin / Reports

Reports & Analytics

Revenue Report

Monthly and daily revenue breakdown with payment method analysis.

Vehicle Report

Most rented vehicles, utilization rates, and revenue per vehicle.

Customer Report

Top customers, spending analysis, and activity trends.

Maintenance Report

Maintenance costs, service history, and upcoming schedules.

14. Future Improvements

14.1 Payment Gateway Integration

Integrate real payment gateways (Stripe, PayPal, bKash) for automated payment processing instead of manual payment tracking.

14.2 Eloquent Model Migration

Create Eloquent models for all domain entities (Vehicle, Booking, Payment, Review, etc.) to leverage relationships, scopes, accessors, and model events.

14.3 REST API Development

Build a RESTful API layer for mobile app integration and third-party service connectivity.

14.4 Real-Time Notifications

Implement WebSocket-based real-time notifications using Laravel Echo and Pusher for instant booking status updates.

14.5 Email Notifications

Configure SMTP/Mailtrap for actual email notifications instead of log-based mail driver.

14.6 Multi-Language Support

Add internationalization (i18n) support for multiple languages using Laravel's localization features.

14.7 Advanced Search

Implement Elasticsearch or Algolia for advanced full-text search with typo tolerance and faceted filtering.

14.8 Mobile Application

Develop a companion mobile application using React Native or Flutter for on-the-go vehicle booking.

14.9 Unit and Feature Tests

Write comprehensive PHPUnit tests for all controllers, models, and business logic to ensure code reliability.

14.10 CI/CD Pipeline

Set up GitHub Actions or similar CI/CD pipeline for automated testing, deployment, and code quality checks.

15. Conclusion

The Vehicle Rental Management System (DriveEase) successfully delivers a comprehensive web-based solution for managing vehicle rental operations. Built on the robust Laravel 12 framework with a SQL architecture, the system provides a complete suite of features for both administrators and customers.

The project demonstrates proficiency in modern web development practices including role based access control, database design with foreign key constraints, transaction-wrapped operations, responsive UI design with Bootstrap 5, and integration with third-party APIs (OpenStreetMap Nominatim for geolocation). The 48 Blade templates, 18 controllers, 16 database tables, and 70 routes form a well-organized and maintainable codebase.

Key achievements include the booking system with date overlap conflict detection, transaction-wrapped approval workflow, comprehensive reporting with PDF export, real-time notification system, and a modern responsive interface. The system handles the complete lifecycle from vehicle browsing through booking, payment tracking, and post-rental review.

The project provides a solid foundation that can be extended with payment gateway integration, Eloquent model migration, REST API development, and mobile application support. The modular architecture and clean separation of concerns make these future enhancements straightforward to implement.